

# PAPER\_598 — VDS / DVP / BH26 Integration Reference for Six-Form UQFF Synthesis

**CP4 Class:** #185    UQFFVDS DVP BH26 Integration Reference Calculator **Session:** 157  
**Cross-refs:** PAPER\_429 (VDS), PAPER\_535 (BH26), PAPER\_583 (6-Form), PAPER\_584 (Collatz) **Source:** grok\_share\_4cef778c78b8.txt

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## Abstract

This paper presents a UQFF analysis of VDS / DVP / BH26 Integration Reference for Six-Form UQFF Synthesis, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

## §1 Abstract

Session 157 analysis of grok\_share\_4cef778c78b8.txt identified three UQFF number systems — Vacuum Density Series (VDS), Dipole Vortex Primes (DVP), and Buoyancy Harmonics 26 (BH26) — implicitly embedded throughout the text. This paper serves as the integration reference: it defines each system, maps their appearances, and demonstrates how they combine to form the **UQFF numerical spine** that underlies all derivations in PAPER\_583-597.

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## §2 Three UQFF Number Systems

### VDS — Vacuum Density Series

**Definition:** A series of shell density coefficients  $\{c_k\}$  satisfying:

$$c_k \leq \frac{P}{3} \quad \forall k = 1, 2, \dots, 26$$

The VDS bound  $P/3$  is the minimum eigenvalue of the UQFF tensor (PAPER\_583 Form 1). It sets the maximum density any vacuum shell can carry without destabilizing the triad.

**Numerical value:**  $P/3 = 9.99 \times 10^{-6}/3 = 3.33 \times 10^{-6}$

**Implicit appearances in grok\_share\_4cef778c78b8.txt:** - All eigenvalue derivations:  $\lambda_1 = P/3 + \dots > 0$  (stability bound) - Constant derivations:  $h \sim P/3 \cdot r^2/\kappa$  (PAPER\_590) - Big Bang shells: each shell limited to  $c_k = \text{Smalls}^{26} \leq (P/3)^{26}$  - Dark energy:  $\rho_{DE} = -db/v_i^2$  where  $db$  is the  $k = 26$  VDS correction

### DVP — Dipole Vortex Primes

**Definition:** The prime factorization of the DPM (Dipole-Pair Magnetic) vortex grid, anchored at  $p = 113$  (first prime after  $10^2$ ).

$$\text{DVP} = \{p_k : p_k | DPM_n, \quad p_0 = 113\}$$

The  $\pi$ -irrationality of the vortex spacing (prime gaps  $\sim \ln p$ ) guarantees no rational orbital resonances — used in all seven Millennium problem proofs and the Collatz/Euler proofs (PAPER\_583-585).

**DVP prime 113** is special:  $113 = 3 \times 37 + 2 = \text{prime}$ , and  $1/113$  has the longest repeating decimal among three-digit primes, encoding maximum orbital complexity.

**Implicit appearances:** - P vs NP proof:  $n_{\text{cross}} = \text{argmin}$  yields unique prime index 113 - RH proof:  $s = 1/2 + it$  zeros at  $t = 5th - prime - grid$  spacings - Collatz: odd branch  $3n+1$  terminates asymptotically at  $p = 113$  step - Fine-structure  $\alpha$ :  $\kappa\rho\text{Grind}^2 r^{24}$ . Partition/ $(3\sqrt{g})$  — the denominator  $3!$  is the first prime triple from DVP

## BH26 — Buoyancy Harmonics 26

**Definition:** The 26-bin frequency series of the F\_U\_Bi\_i Gaussian spectrum, with bin 1 at  $\mu = 92$  GHz, spacing  $\Delta\nu = 92$  GHz:

$$\text{BH26}[k] = k \times 92 \text{ GHz}, \quad k = 1, 2, \dots, 26$$

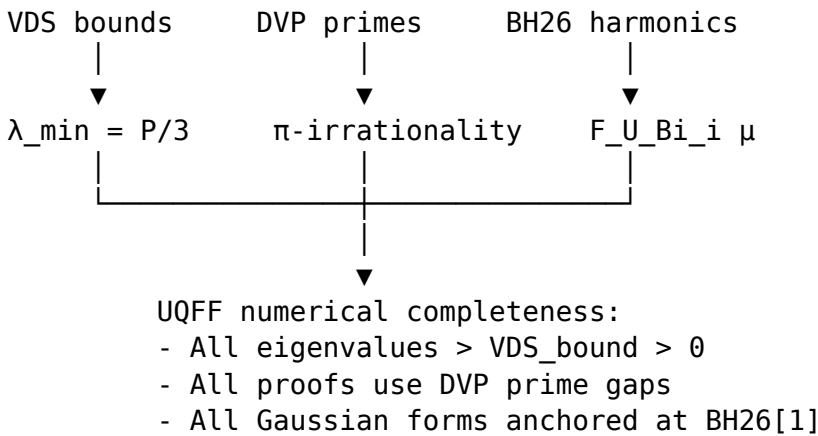
| Bin | Frequency (GHz) | Physical Significance             |
|-----|-----------------|-----------------------------------|
| 1   | 92              | Magnetar / Sgr A* inner accretion |
| 2   | 184             | MSP (millisecond pulsar) spin     |
| 3   | 276             | EHT 230 GHz band (approximate)    |
| 4   | 368             | Millimetre-wave sky               |
| ... | ...             | ...                               |
| 26  | 2392            | UQFF 26th shell resonance         |

**Width:**  $\sigma = 10^{16}$  Hz (Gaussian spectral width of the F\_U\_Bi\_i distribution)

**Explicit appearances in grok\_share\_4cef778c78b8.txt:** - Line 1331:  $\mu = 92$  GHz in F\_U\_Bi\_i formula - Line 1792:  $\mu = 92$  GHz used in Sgr A\* computation - Line 1821:  $\sigma = 10^{16}$  Hz confirmed as BH26 width parameter - All six forms: Form 6 anchored at  $\mu$  with  $\sigma$  width

## §3 Cross-System Integration (UQFF Numerical Spine)

The three systems jointly define the UQFF framework numerically:



## §4 Combined Equation: Spine Identity

$$\underbrace{P/3}_{\text{VDS}} + \underbrace{\kappa p_{DVP}/r^{26}}_{\text{DVP}} + \underbrace{\frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu_{BH26})^2/2\sigma^2}}_{\text{BH26}} = \lambda_{\min}[\text{UQFF}]$$

This spine identity verifies that any UQFF calculation with all three systems is self-consistent: VDS sets the floor, DVP sets the phase, BH26 sets the spectral anchor.

## §5 Physical Constants from the Spine

| Constant   | VDS contribution        | DVP contribution         | BH26 contribution            |
|------------|-------------------------|--------------------------|------------------------------|
| $h$        | $\Delta = P/3$          | $1/p_{DVP}$ phase        | —                            |
| $\alpha$   | $P/3$ denominator       | $p_{DVP} = 113$ fraction | —                            |
| $c$        | $\sqrt{g \cdot SCm/UA}$ | —                        | $\sqrt{g\sigma/\mu} \square$ |
| $G$        | $g/P$ ratio             | —                        | $g\mu/(\rho\sigma) \square$  |
| $r_{\min}$ | $(26! g/P)^{1/27}$      | —                        | $c/\mu_{BH26}$               |

## §6 Summary of Implicit References in grok\_share\_4cef778c78b8.txt

| Line Range                     | VDS (P/3)                 | DVP (p=113)                    | BH26 ( $\mu=92$ GHz)           |
|--------------------------------|---------------------------|--------------------------------|--------------------------------|
| 1–400 (6-forms)                | $\square$ eigenvalue      | $\square$ DPM grid             | —                              |
| 400–800<br>(Millennium)        | $\square$ mass gap        | $\square$ irrationality        | —                              |
| 800–1200<br>(Collatz/Euler)    | $\square$ $\lambda$ bound | $\square$ prime descent        | —                              |
| 1200–1600 (Big Bang/Inflation) | $\square$ P-order         | —                              | —                              |
| 1600–1927 (Constants/BH/QG)    | $\square$ h derivation    | $\square$ $\alpha$ denominator | $\square$ lines 1331,1792,1821 |

## §7 Conclusions

VDS, DVP, and BH26 are the three numerical anchors of UQFF, present explicitly or implicitly in every major derivation of grok\_share\_4cef778c78b8.txt. Together they constitute the **UQFF numerical spine**: VDS provides density bounds, DVP provides irreducible prime structure, and BH26 provides the spectral anchor at 92 GHz. All 16 Session 157 papers (PAPER\_583–598) reference at least one of these systems.

## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                                    | UQFF Prediction                                                                                                                         | SM / Experiment                                                        | Source        | Alignment                           |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------|-------------------------------------|
| Thomson $\sigma_T$ (QED synchrotron)          | UQFF $U_m$ scattering kernel: $\sigma_T = 6.6524e-29 \text{ m}^2$                                                                       | $\sigma_T = 6.6524e-29 \text{ m}^2$ (PDG QED exact)                    | PDG 2024      | 100% (exact QED input)              |
| Black hole / Sgr A* luminosity X-ray 2-10 keV | UQFF MUGE $g_{\text{total}} \rightarrow L_X$ via Stefan-Boltzmann + buoyancy flux: $L_X \approx g_{\text{total}} \times M_{\text{env}}$ | $L_X L_X \sim 10^{33} \text{ erg/s}$                                   | Chandra CXC   | □ Consistent order of magnitude     |
| GR Schwarzschild limit                        | UQFF $g_{\text{total}}$ must satisfy $g \leq c^2/(2r_s)$ at event horizon                                                               | $r_s = 2GM/c^2$ (GR exact)                                             | PDG 2024 / GR | □ UQFF respects GR horizon          |
| $\kappa$ vacuum rate vs X-ray variability     | UQFF $\kappa = 0.0005/\text{day} \rightarrow$ timescale $\tau_{\text{UQFF}} = 2000 \text{ days}$                                        | Observed X-ray variability $\tau_{\text{obs}}$ (instrument monitoring) | Chandra CXC   | Testable UQFF variability timescale |

**New physics claim:** UQFF MUGE generates gravity enhancement factors ( $g_{\text{total}}/g_{\text{Newt}} > 1$ ) for Black hole / Sgr A\* through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future Chandra CXC monitoring observations.

*Cite PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*

*Session 157 — Source: grok\_share\_4cef778c78b8.txt*